

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Section 20

Acceptance

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1. GENERAL INFORMATION

1.1. GENERAL

All scans from the technical image Quality Test can (and should) be performed in batch-mode (individually or grouped) using SPT. This has to be done for the following reasons:

- Correct setting of control parameters (uniform scaling, uniformity correction for the bodycoil, filtersetting, etc)
- Correct setting of protocol parameters
- Automatic evaluation of images
- Compare with specifications
- Enabling of the electronic load

NOTE

1.0T and 1.5T Systems.

Make sure that the Ehrhorn amplifier has been switched on for at least 15 minutes before starting the performance measurements.

1.2. PROCEDURE TO PERFORM MEASUREMENTS

In this paragraph the general procedure for using SPT to perform a scan is given:

- Select : Scan control.
- Select : SPT scanning.
- Select : Service mode.
- Select : Perform IQP/EST measurements.
- Select : Analysis principle. Select the default: IQP.
- Select : Coil type.
- Select : Scan orientation (not always applicable).

If more scans are available, a list of all the scans is given on the screen.
Select one or more scans.

- Position the phantom according to the instructions given on the screen.
- Press <PROCEED> to execute the selected scans.

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1.3. PROCEDURE TO CREATE AN MRL

Before an MRL can be made, two spec-files must be downloaded from the BBS and installed onto the system. Refer to section 22 'MRL', paragraph 7 'image quality specifications' for more information regarding the spec-files.

The general procedure to view/print the results of the automatic evaluation is as follows:

- Select : ICON 'System Performance Tools'
- Select : Service mode
- Select : System verification
- Select : Measurement result lists
A list of available data files is provided. Format: SPT_* (in which the * stands for the examination name)
- Select : One or more of the available files.
- Select : The type of the MRL
- Select : Sections to be shown
- Select : Output device(s)

2. PRE-PERFORMANCE TEST

The purpose of the pre-performance test described in this section is to ensure that the image quality test reveals no major/obvious IQ problems. If the system does not meet the requirements of these test, it is of no use to perform the image quality test. The whole system must be operational and all parameters must have the correct value.

The following test should be performed before starting the image quality test:

- artifact measurements
- spike testbatch
- PIQT

The following test is optional:

- connector test

2.1. ARTIFACT MEASUREMENTS

For the artifact measurements with the headcoil the 250 cc bottle is used.

For the measurements with the bodycoil the 250 cc bottle may be used. If you are scanning with the bodycoil on the 250 cc bottle and you get the message 'Cannot determine optimum RF Power', use the 1000 cc bottle instead of the 250 cc bottle.

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2.1.1. ARTIFACT MEASUREMENT PROCEDURE

The artifact measurements are specified at different locations in the magnet. These locations with their corresponding procedures are shown in the next table:

Protocol:	H01	H02	B01	B02	B03
Coil type	Head		Body		
Scan orientation	T		T	C	S
Bottle orientation	axis in z-dir.		axis in z-dir.	axis in y-dir.	axis in x-dir.
Co-ordinates of position ¹	-5,-5,0		-11,-11,0	0,-11,-11	-11,0,-11

Table 1: Bottle location artifact measurements.

The artifact measurement is performed under normal conditions (IQP).

For the bodycoil the procedure is:

- Select : Scan control.
- Select : SPT Scanning.
- Select : Service mode.
- Select : Perform IQP/EST measurements.
- Select : Analysis principle EST.
- Select : Body_Quad
- Select : Select the required 'scan orientation'

The list of available scans is:

- 1 B0*_ES
- 2 B0*_IQ

In which * corresponds to 1, 2, 3 for respectively the Transversal, Coronal and Sagittal scan orientation.

- Select : 2
- After positioning the bottle in the bodycoil (see table 1) press <PROCEED>

For the headcoil the procedure is:

-
- Select : Analysis principle EST.
- Select : Head_Quad

The list of available scans is:

- 1 H01_ES
- 2 H01_IQ
- 3 H02_ES
- 4 H02_IQ

- Select 2 and 4
- After positioning the bottle in the headcoil (see table 1) press <PROCEED>

¹ X (Horizontal), Y (Vertical), Z (Axial).

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2.1.2. PROCEDURE TO EVALUATE THE ARTIFACT MEASUREMENTS.

First the patient (bottle) which contains the artifact measurements must be selected.

Procedure:

- Select patient
- Select Examination
- Select the 'Bottle' examination.
- Select the scan which you want to analyse.

The intensity of ghost images in a modulus image should be measured as follows:

- Scale the image in such a way that the maximum ghost becomes visible.
- Position the arrow in the image and press the right mouse button.
- Select MORE and ROI Settings.
- Select the square ICON and HIDE.
- Draw a square ROI.
- The ROI can be moved over the image by putting the arrow in the middle of the ROI and pressing the left mouse button.
- Select the number above the ROI using the right mouse button. Select STATISTICS and again STATISTICS.
- A window will appear in which the mean value and the ROI size in pixels is given.
- Put the arrow in a corner of the ROI and rescale the ROI until the 'total pixel area' is about 49 mm².
- Measure the mean intensity S at the centre of the bottle.
- Measure the mean intensity G at the spot where the maximum intensity ghost is seen.
- Measure the mean intensity B and the standard deviation sd(B) of a ghost free part of the background of the image.
In a ghost-free part of the background the value B/sd(B) lies in the range: $1.8 < B/sd(B) < 2.2$.
- The intensity of the ghost must be calculated using the relation:

$$\text{ghost - intensity} = \left(\frac{G - B}{S} \right) * 100\%$$

- Write the measured values in the MRL.

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2.2. SPIKE TESTBATCH.

2.2.1. GENERAL SPIKE INFORMATION.

Spikes are visible as white dots in the raw data. Depending on the intensity and the number of spikes, the effects in the reconstructed image vary from some reduction in the signal to noise ratio to a pattern of interference lines in the image. Spikes can be caused by 'poor' mechanical contacts, electrostatic discharges and conducting objects which vibrate against another conducting object. Since the Power and Master gradient chains use stronger gradients, spikes are more likely with these configurations.

It is advised to check the following items before starting the performance measurements:

- All connections in the gradient chain (coil, interconnection strip, filterbox and amplifier).
- All connectors on the coilint, detune driver, PFEI NL, Filterbox, NTDAC.
- All ground connections.
- Remove the ladder from the magnet before starting the performance measurements. Advise is not to store the ladder with the magnet but in the technical room or another practical place.

The following items are also known as potential spike sources. However, since they are rather difficult to access or can only be checked by performing a scan, they only have to be checked when spikes occur:

- Detune boards and other mechanical/electrical connections inside the bodycoil.
- Connections inside the hybrid box.
- Surface coil connectors (try different connector!).
- Movement (vibration) of airducts and quench pipes against other conducting objects.
- Electrical discharge of the false ceiling (static electricity due to airflow).
- Static electricity caused by (gradient) cable movement.
- Emergency light in the magnet room.
- Smoke detectors in the examination room.

2.2.2. IMAGING SPIKE TESTBATCH.

An imaging spike testbatch must be executed before starting the performance measurements. This is to check if the spike level of the system is within acceptable limits. This test will perform a number of heavy scans. During the reconstruction, the raw-data of the scans is checked for spikes. The measured number and amplitude of spikes are translated into a QPI factor. It must be checked on completion of the scan that the QPI of each scan is smaller than 1. Depending on the availability of the required options, some scans may not be performed during the execution of the batch. When the required options are not available, the warning 'conflicting parameters' and 'define scan failed' is generated and the batch will continue with the next scan.

2.2.3. RUNNING THE TESTBATCH.

Procedure:

- Prepare the quad headcoil and put the 1 litre bottle horizontally in the middle of the coil.
- Use the light-visor and travel-to-scanplane button to position the headcoil and bottle in the isocentre of the magnet.
- Startup : GYROSCAN
- Select : Scan Control
- Select : Scan utilities

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Select : Enter service mode
 Select : Batch interpreter
 Select : Testbatch tb_spptx000
 (Standard and Omni x = 1, Power x = 3, Master x = 6).

2.2.4. ANALYZING THE RESULTS.

The following procedure can be used to analyze the results:

- 1) Select: The 'Acquisition Context' icon.
- 2) Select: Prev. Scans
- 3) Select: Display parameters
- 4) Select: Examination 'Spike scans'
- 5) Select: A scan of the spike testbatch.
- 6) Select: System performance tool parameters
- 7) Select: Check the parameter 'SP_JDB_QPI' (last page). The value should be less than 1.
- 8) Select: <cancel> 2x
- 9) Repeat step 5 - 8 until the QPI for the different scans has been checked.

2.3. CONNECTOR TEST (OPTIONAL).

2.3.1. PROCEDURE FOR OPERATION.

- Use the procedure given in paragraph 1.2 to select the batch files available for the 'Head-quad coil'.
- Select scan 1 (CT1:CON_TEST) and connect the headcoil to connector 1.
- Press <PROCEED> to perform the scan.
- After the scans for connector 1 have completed, the headcoil can be connected to connector 2.
- Press <PROCEED> to start the second scan.

2.3.2. CHECKING THE RESULTS

The following procedure can be used to check the connector test results

- Select: ICON 'System performance tools'.
- Select: IQ parameter/data file evaluation tools
- Select: SNR and Flood Field Uniformity measurements
- Select: Examination 'CONNECTOR'
- Select: The scan made with connector 1.
- Select: Slice 3.
- Select: Philips quality procedure.

After entering proceed, a 'flood field uniformity' analysis is performed and the results are displayed on the screen. Write down the SNR(B) Value.

- Select: The 'connector 2' scan.
- Select: Philips quality procedure.

After entering proceed, a 'flood field uniformity' analysis is performed and the results are displayed on the screen.

Write down the SNR(B) Value.

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- Calculate the 'deviation' of the SNR(B) of each connector compared with the average SNR(B) value using the next formula:

$$deviation[n] = \frac{|SNR(B)[n] - average|}{average}, average = \frac{SNR(B)[1] + SNR(B)[2]}{2}$$

In which 'n' stands for the number of the used connector.

- Write the measured values in the MRL.

2.4. PERIODIC IMAGE QUALITY TEST

This test consists of three transversal scans using the 200 mm performance phantom. For the first two scans (QA1 and QA2) the headcoil is used while for the third scan (QA3) the bodycoil is used as a receive coil. Online information concerning PIQT is available via menu item 4, available after pressing function key <SPT scanning>.

2.4.1. PROCEDURE FOR OPERATION

- Select : Scan control.
- Select : SPT Scanning.
- Select : PIQT.
- Position the phantom according to the procedure given on the screen.
- After pressing <PROCEED> the scans will be executed automatically.

2.4.2. PROCEDURE FOR EVALUATION

The images are analyzed automatically. The MRL can be created as described in 1.3.

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3. PERFORMANCE MEASUREMENTS (HEAD/BODY AND SURFACE COILS).

3.1. PROCEDURE FOR OPERATION

The scans for the technical image quality test should be performed with 'perform measurements' using SPT (see paragraph 1.2). The 'analysis principle' should be set to IQP.

Note that in principle it is sufficient to run only the transversal batch.

After selecting the required coil and scan orientation a selection can be made of the different scans available in the batch. Depending on the configuration and the installed software options, some scans cannot be executed and therefore should not be selected.

HeadT:	The connector test can be skipped. Scan H36: (TSE) can only be executed when the TSE option is installed. Scan H38: (EPI) can only be executed when the Real Time option is installed.
HeadS:	Scan H66: (TSE) can only be executed when the TSE option is installed. Scan H68: (EPI) can only be executed when the Real Time option is installed.
HeadC:	Scan H96: (TSE) can only be executed when the TSE option is installed. Scan H98: (EPI) can only be executed when the Real Time option is installed.
BodyT:	Scan B36: (TSE) can only be executed when the TSE option is installed. Scan B38: (EPI) can only be executed when the Real Time option is installed.
BodyS:	Scan B66: (TSE) can only be executed when the TSE option is installed. Scan B68: (EPI) can only be executed when the Real Time option is installed.
BodyC:	Scan B96: (TSE) can only be executed when the TSE option is installed. Scan B98: (EPI) can only be executed when the Real Time option is installed.

All other scans can be executed with the basic software.

An overview of the available software options can be found in the Gyrotool account.

Procedure:

- Startup : Gyrotool
- Select : Modify system configuration
- Select : Options (SW-Keys).
- Select : View SW options, modify option state

After selecting a certain batch, a helptext (and for the body phantom also an image) is given to assist the user to position both the coil and phantom correctly. The text and images which are given just before the start of the scans, are also given in paragraph 3.2.2.

The scans are executed after pressing <PROCEED>.

Scouts scans are done to obtain the correct values for the offcentre distances and angulations. It is important that the phantom is positioned correctly because otherwise an error message is given and no further scans are executed.

The analysis of the images is done automatically; the results are given in data files which can be recognized by SPT_HEAD* and SPT_BODY* for the headcoil and bodycoil respectively. The * stands for the orientation (T, S or C).

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Standard	Omni	Power	Omni	Power	Master

3.2. PHANTOM POSITIONING

3.2.1. THE 200 MM PERFORMANCE PHANTOM

The positioning of the holder and phantom must be executed as follows:

Lightbeams:

The lightbeams are used to centre the headcoil in the middle of the magnet. Press the <LIGHT> button on the patient control panel to activate the lightbeams. The lightbeams form a cross pattern which defines the transversal and sagittal planes.

Move the table in or out, until the transversal lightbeam line coincides with the markers on the headcoil. Press the Travel-to-scanplane button and use the Tumble switch to move the phantom to the 'scan plane'.

Holder:

Remove the headrest from the headcoil and place the head phantom holder inside the coil.

The innerside of the holder is asymmetric which corresponds to the notches of the head phantom (see Figure 4).

Phantom:

The head phantom has two notches which are used to secure the position of the phantom for the three different orientations.

For every direction the capital ('T', 'S' or 'C') which corresponds to the orientation to be scanned should be visible at the frontside. In that case the notch will secure the phantom in that orientation.

Intera 0.5 T			Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master	

3.2.2. THE 400 MM PERFORMANCE PHANTOM

The positioning of the holder and phantom must be executed as follows:

holder:

Align the table in such a way that the lightbeam hits the outmost end of the table top.

Use the function 'TRAVEL-TO-SCANPLANE' till the end of the tabletop is in the scanplane position. Now move it back over a distance of 210 ± 2 mm.

Place the phantom holder with phantom in the magnet and push it gently till it touches the table top.

Phantom transversal:

- For correct position see Figure 1
- The spatial linearity section should point to the frontside of the magnet (when the phantom is positioned in the magnet).
- Make sure that the markerline is horizontal (line can be seen from the rearside of the magnet).

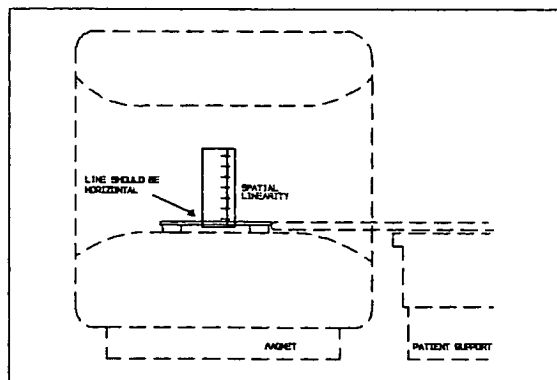


Figure 1 Transversal position of phantom

Phantom sagittal:

- For correct position see Figure 2
- The spatial linearity section should point to the rightside of the magnet (seen from the frontside of the magnet).
- Make sure that the markerline is horizontal.

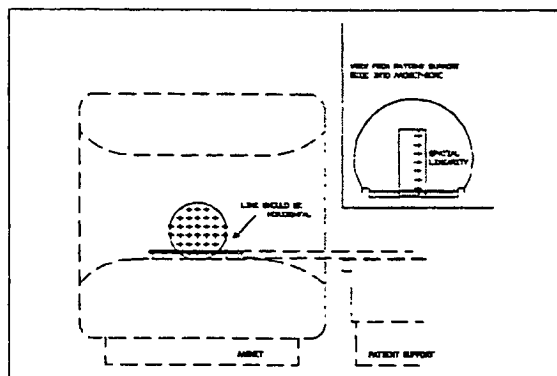


Figure 2 Sagittal position of phantom

Phantom coronal:

- For correct position see Figure 3
- The spatial linearity section should point to the bottom of the magnet.
- Make sure that the markerline is parallel to the z-axis.

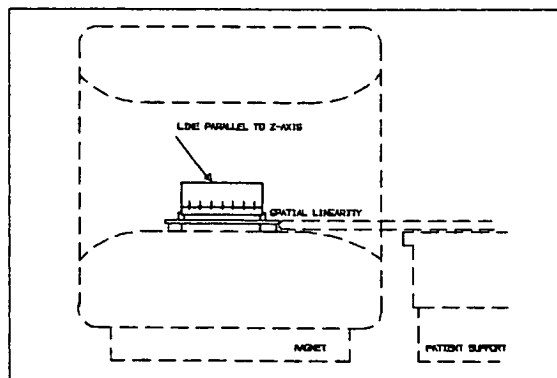


Figure 3 Coronal position of phantom

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4. PHANTOM SET

4.1. INTRODUCTION

For the GYROSCAN the Philips phantom set contains the following phantoms:

- 250 cc bottle
- 1000 cc bottle
- 3000 cc bottle
- 5000 cc bottle
- 200 mm performance phantom + holder (including the load segment)
- 400 mm performance phantom + holder

In general the phantoms are made of Poly Methyl Methane Acrylate (PMMA). The accuracy of the relaxation times of the phantom fluids is $\pm 5\%$.

The standard phantom fluids with their specific relaxation times are summarized in the following table.

CONTENTS:	T1 (msec)			T2 (msec)		
	0.5T	1.0T	1.5T	0.5T	1.0T	1.5T
FLUID 1: 1 litre demi water + 770 mg $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ + 1 ml arquad (1% solution) + 0.15 ml H_2SO_4 - 0.1N solution	300	330	360	280	300	320
FLUID 2: FLUID 1 + 2 gr/l NaCl	300	330	360	280	300	280

Table 2: Relaxation times phantom fluid.

4.2. BOTTLE PHANTOMS

All the bottles are filled with phantom fluid 2.

The bottles are used for different purposes such as:

1. measure the artifact level
2. measure the Image Quality of some surface coils
3. obtain a load for some other surface coils.
4. determine parameters using System Tuning Tools

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
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4.3. THE 200 MM PERFORMANCE PHANTOM

The 200 mm performance phantom is a cylinder of plexiglas having an internal diameter of 200 millimetres and a length of 110 millimetres. The holder can be clicked on in the headcoil, and is designed with positioning steps and pins (see Figure 4).

Together with the shape of the phantom, which is given by two notches, this is used to secure the position of the phantom in only three directions: transversal, sagittal and coronal. Every direction of the phantom is marked with a capital ('T', 'S' or 'C') which corresponds with the orientation to be scanned. The capital should point to the frontside of the table.

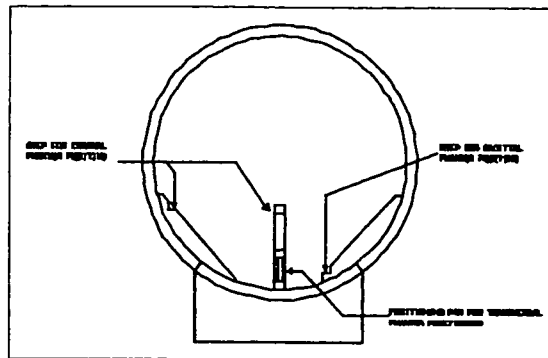


Figure 4 Holder for headphantom

The phantom contains a series of structures at approximately equidistant axial positions for measuring the important Image Quality characteristics.

The phantom is filled with FLUID 1 and contains the following four inserts (see Figure 5):

1. Spatial resolution / ghost
2. Slice profile
3. Floodfield uniformity
4. Spatial linearity

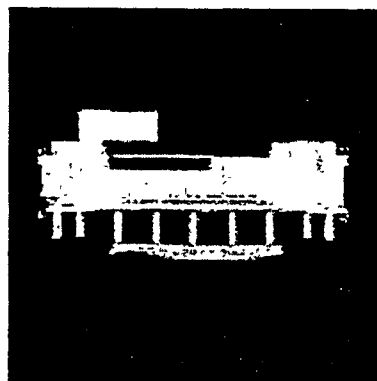


Figure 5 Overview of headphantom

4.3.1. SPATIAL RESOLUTION/GHOST SECTION

This section (see Figure 6) is used to measure the spatial resolution of the system.

The insert is made of a 20 mm thick disk, in which a large square hole is present. The dimension of the hole is 50x50 mm, while the rotation is 11.3 degrees.

The spatial resolution is measured in both the measurement and preparation direction. The distance between the centre of the hole and the centre of the section is 50 mm.

This section can also be used for evaluation of the artifact level.

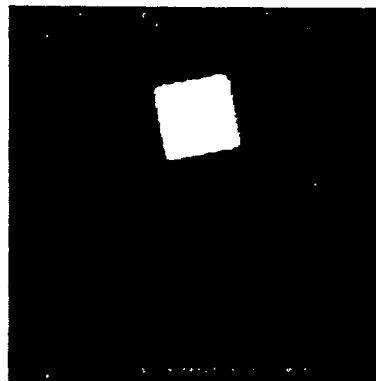


Figure 6 Spatial resolution

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4.3.2. SLICE PROFILE SECTION

This section (see Figure 7) is used to measure the slice profile of the system

This insert consists of two parts:

- a double ramp with a width of 1 mm
- a wedge

The angle for both the ramps and the wedge is 11.3 ± 0.17 degrees.

The wedge is used for slices with a thickness of less than 5 mm, otherwise the ramp is used.

The double ramp is also used to determine the exact angle of the ramp with respect to the scanplane.

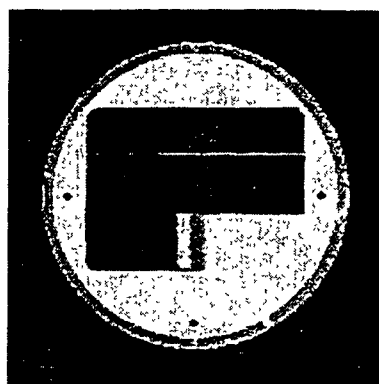


Figure 7 Slice profile

4.3.3. FLOODFIELD UNIFORMITY SECTION

This section (see Figure 8) is used to measure the floodfield uniformity and signal to noise ratio.

Also the artifact level is determined (i.e. the maximum intensity, within a small ROI, with respect to the intensity in the centre).

All these Image Quality parameters can be derived from this section, which contains the phantom FLUID 1.

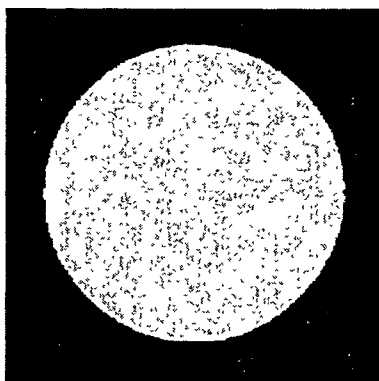


Figure 8 Floodfield uniformity

4.3.4. SPATIAL LINEARITY SECTION

This section (see Figure 9) is used to measure the spatial linearity of the system.

This insert consists of a square array of holes.

The diameter of the holes is 5 mm while the distance between the centres of the holes is 25 mm.

This section can also be used to adjust the gradient strength by measuring the distance between the outer holes in both the horizontal and vertical direction.

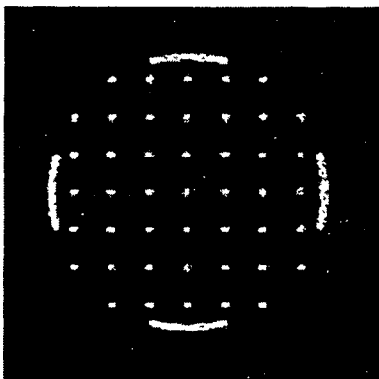


Figure 9 Spatial linearity

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4.4. THE 400 MM PERFORMANCE PHANTOM

The 400 mm performance phantom is a cylinder of plexiglas having an external diameter of 400 millimetres and a length of 100 millimetres.

The phantom is filled with FLUID 1 and contains the following two sections (see Figure 10): Spatial linearity and Floodfield uniformity

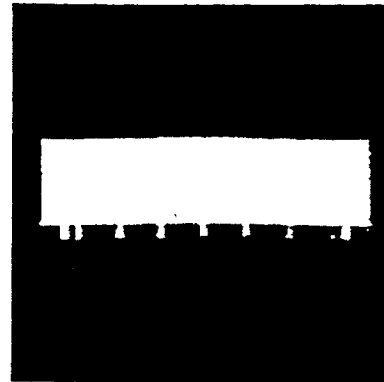


Figure 10 Overview of bodyphantom

4.4.1. SPATIAL LINEARITY SECTION

This section (see Figure 11) is used to measure the spatial linearity of the system.

This insert consists of a square array of holes. The diameter of the holes is 10 mm while the distance between the centres of the holes is 50 mm.

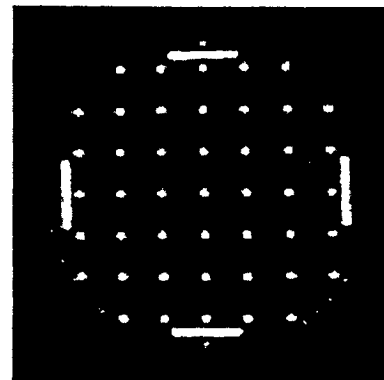


Figure 11 Spatial linearity